

Creation Date 29-Jan-2010

Revision Date 06-Oct-2023

Revision Number 11

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                                  |                                                              |
|----------------------------------|--------------------------------------------------------------|
| <b>Product Description:</b>      | <b>Sodium borohydride caplets</b>                            |
| <b>Cat No. :</b>                 | <b>448490000; 448490025; 448490100; 448491000; 448495000</b> |
| <b>Synonyms</b>                  | SBH; Sodium tetrahydroborate                                 |
| <b>CAS No</b>                    | 16940-66-2                                                   |
| <b>EC No</b>                     | 241-004-4                                                    |
| <b>Molecular Formula</b>         | H4 B Na                                                      |
| <b>REACH registration number</b> | 01-2119485016-39                                             |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                       |                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Recommended Use</b>                | Laboratory chemicals.                                                                       |
| <b>Sector of use</b>                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| <b>Product category</b>               | PC21 - Laboratory chemicals                                                                 |
| <b>Process categories</b>             | PROC15 - Use as a laboratory reagent                                                        |
| <b>Environmental release category</b> | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| <b>Uses advised against</b>           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel,  
Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG,  
United Kingdom

**Swiss distributor - Fisher Scientific AG**  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - Regulation (EC) No 1272/2008

##### Physical hazards

Substances/mixtures which, in contact with water, emit flammable gases

Category 1 (H260)

##### Health hazards

Acute oral toxicity

Category 3 (H301)

Skin Corrosion/Irritation

Category 1 C (H314)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Reproductive Toxicity

Category 1B (H360FD)

##### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

### 2.2. Label elements

Contains Sodium borohydride



Signal Word

Danger

#### Hazard Statements

H260 - In contact with water releases flammable gases which may ignite spontaneously

H314 - Causes severe skin burns and eye damage

H301 - Toxic if swallowed

H360FD - May damage fertility. May damage the unborn child

EUH014 - Reacts violently with water

#### Precautionary Statements

P223 - Do not allow contact with water

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

## Additional EU labelling

Restricted to professional users

## 2.3. Other hazards

Reacts violently with water

In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component          | CAS No     | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                          |
|--------------------|------------|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------|
| Sodium borohydride | 16940-66-2 | EEC No. 241-004-4 | >95      | Water-react. 1 (H260)<br>Acute Tox. 3 (H301)<br>Skin Corr. 1C (H314)<br>Eye Dam. 1 (H318)<br>Repr. 1B (H360FD)<br>(EUH014) |

| Component          | Specific concentration limits (SCL's) | M-Factor | Component notes |
|--------------------|---------------------------------------|----------|-----------------|
| Sodium borohydride | >=3.4% Repr. 1B                       | -        | -               |

|                           |                  |
|---------------------------|------------------|
| REACH registration number | 01-2119485016-39 |
|---------------------------|------------------|

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                    |                                                                                                                                                                                                                                                                                                                                |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General Advice                     | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                              |
| Eye Contact                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| Skin Contact                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| Ingestion                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| Inhalation                         | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. |
| Self-Protection of the First Aider | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

## **4.3. Indication of any immediate medical attention and special treatment needed**

Notes to Physician

Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

Water.

### **5.2. Special hazards arising from the substance or mixture**

Corrosive material. Reacts violently with water. Contact with water liberates extremely flammable gases. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

#### **Hazardous Combustion Products**

Oxides of boron, Hydrogen, Thermal decomposition can lead to release of irritating gases and vapors, Sodium oxides.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

### **6.2. Environmental precautions**

Avoid release to the environment. See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Do not expose spill to water. Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Avoid dust formation. Do not get in

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe (dust, vapor, mist, gas). Do not allow contact with water.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Keep away from water or moist air. Do not store in aluminum containers.

## Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany)

Storage Class/LGK 4.3

## Switzerland - Storage of hazardous substances

Storage class - SC 4.3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

#### Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

Workers; See table for values

| Component                              | Acute effects local (Oral) | Acute effects systemic (Oral) | Chronic effects local (Oral) | Chronic effects systemic (Oral) |
|----------------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------------|
| Sodium borohydride<br>16940-66-2 (>95) |                            |                               |                              | 0.17 mg/kg bw/day               |

| Component | Acute effects local | Acute effects | Chronic effects local | Chronic effects |
|-----------|---------------------|---------------|-----------------------|-----------------|
|-----------|---------------------|---------------|-----------------------|-----------------|

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

|                                        | (Dermal) | systemic (Dermal) | (Dermal) | systemic (Dermal)         |
|----------------------------------------|----------|-------------------|----------|---------------------------|
| Sodium borohydride<br>16940-66-2 (>95) |          |                   |          | DNEL = 240mg/kg<br>bw/day |

| Component                              | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation) | Chronic effects<br>systemic (Inhalation) |
|----------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|
| Sodium borohydride<br>16940-66-2 (>95) |                                     |                                        |                                       | DNEL = 5.1mg/m <sup>3</sup>              |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                              | Fresh water     | Fresh water<br>sediment         | Water Intermittent | Microorganisms in<br>sewage treatment | Soil (Agriculture)         |
|----------------------------------------|-----------------|---------------------------------|--------------------|---------------------------------------|----------------------------|
| Sodium borohydride<br>16940-66-2 (>95) | PNEC = 1.75mg/L | PNEC = 2.55mg/kg<br>sediment dw | PNEC = 1.75mg/L    | PNEC = 54.77mg/L                      | PNEC = 4.8mg/kg<br>soil dw |

| Component                              | Marine water    | Marine water<br>sediment            | Marine water<br>Intermittent | Food chain | Air |
|----------------------------------------|-----------------|-------------------------------------|------------------------------|------------|-----|
| Sodium borohydride<br>16940-66-2 (>95) | PNEC = 1.75mg/L | PNEC =<br>0.255mg/kg<br>sediment dw |                              |            |     |

## 8.2. Exposure controls

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                                                  |                                          |
|------------------------------------------------|------------------------------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Solid Powder                                                     |                                          |
| <b>Appearance</b>                              | White                                                            |                                          |
| <b>Odor</b>                                    | Odorless                                                         |                                          |
| <b>Odor Threshold</b>                          | No data available                                                |                                          |
| <b>Melting Point/Range</b>                     | 360 °C / 680 °F                                                  |                                          |
| <b>Softening Point</b>                         | No data available                                                |                                          |
| <b>Boiling Point/Range</b>                     | No information available                                         |                                          |
| <b>Flammability (liquid)</b>                   | Not applicable                                                   | Solid                                    |
| <b>Flammability (solid,gas)</b>                | No information available                                         |                                          |
| <b>Explosion Limits</b>                        | No data available                                                |                                          |
| <b>Flash Point</b>                             | No information available                                         | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 220 °C / 428 °F                                                  |                                          |
| <b>Decomposition Temperature</b>               | 400 °C                                                           |                                          |
| <b>pH</b>                                      | approx 11                                                        | 10 g/l aq.solution                       |
| <b>Viscosity</b>                               | Not applicable                                                   | Solid                                    |
| <b>Water Solubility</b>                        | Reacts violently with water                                      |                                          |
| <b>Solubility in other solvents</b>            | No information available                                         |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                                  |                                          |
| <b>Vapor Pressure</b>                          | negligible                                                       |                                          |
| <b>Density / Specific Gravity</b>              | 1.074                                                            |                                          |
| <b>Bulk Density</b>                            | powder: 400 kg/m <sup>3</sup><br>granules: 510 kg/m <sup>3</sup> |                                          |
| <b>Vapor Density</b>                           | Not applicable                                                   | Solid                                    |
| <b>Particle characteristics</b>                | No data available                                                |                                          |

### 9.2. Other information

|                                                                               |                                   |                    |
|-------------------------------------------------------------------------------|-----------------------------------|--------------------|
| <b>Molecular Formula</b>                                                      | H4 B Na                           |                    |
| <b>Molecular Weight</b>                                                       | 37.83                             |                    |
| <b>Substances/mixtures which, in contact with water, emit flammable gases</b> | Emitted gas ignites spontaneously | Gas(es) = Hydrogen |
| <b>Oxidizing Properties</b>                                                   | Not oxidising                     |                    |
| <b>Evaporation Rate</b>                                                       | Not applicable - Solid            |                    |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Water reactive. Hygroscopic.

### 10.3. Possibility of hazardous reactions

**Hazardous Polymerization** Hazardous polymerization does not occur.

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

## Hazardous Reactions

Contact with water liberates extremely flammable gases.

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture. Temperatures above 60°C.

### 10.5. Incompatible materials

Strong oxidizing agents. Aldehydes. Ketones. Acids. Aluminium.

### 10.6. Hazardous decomposition products

Oxides of boron. Hydrogen. Thermal decomposition can lead to release of irritating gases and vapors. Sodium oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Category 3

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component          | LD50 Oral        | LD50 Dermal            | LC50 Inhalation              |
|--------------------|------------------|------------------------|------------------------------|
| Sodium borohydride | 57 mg/kg ( Rat ) | >2000 mg/kg ( Rabbit ) | LC50 > 5.18 mg/L ( Rat ) 1 h |

##### (b) skin corrosion/irritation;

Category 1 C

##### (c) serious eye damage/irritation;

Category 1

##### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

##### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

##### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

Category 1B

##### (h) STOT-single exposure;

Based on available data, the classification criteria are not met

##### (i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

##### (j) aspiration hazard;

Not applicable  
Solid



# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

**Symptoms / effects, both acute and delayed** Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

**Ecotoxicity effects** Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available.

### 12.2. Persistence and degradability

**Persistence** Persistence is unlikely, based on information available.  
**Degradability** Not relevant for inorganic substances, Reacts with water.  
**Degradation in sewage treatment plant** Water reactive. Reacts violently with water.

### 12.3. Bioaccumulative potential

Product does not bioaccumulate due to reaction with water

### 12.4. Mobility in soil

Reacts with water. Reacts violently with water. Is not likely mobile in the environment.

### 12.5. Results of PBT and vPvB assessment

Reacts violently with water. In accordance with Annex XIII of the REACH Regulation, inorganic substances do not require assessment.

### 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

### 12.7. Other adverse effects

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

## Other Information

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

## Switzerland - Waste Ordinance

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

**14.1. UN number** UN1426  
**14.2. UN proper shipping name** SODIUM BOROHYDRIDE  
**14.3. Transport hazard class(es)** 4.3  
**14.4. Packing group** I

### ADR

**14.1. UN number** UN1426  
**14.2. UN proper shipping name** SODIUM BOROHYDRIDE  
**14.3. Transport hazard class(es)** 4.3  
**14.4. Packing group** I

### IATA

**14.1. UN number** UN1426  
**14.2. UN proper shipping name** SODIUM BOROHYDRIDE  
**14.3. Transport hazard class(es)** 4.3  
**14.4. Packing group** I

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component          | CAS No     | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------------|------------|-----------|--------|-----|-------|------|----------|------|------|
| Sodium borohydride | 16940-66-2 | 241-004-4 | -      | -   | X     | X    | KE-31365 | X    | X    |

| Component          | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Sodium borohydride | 16940-66-2 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Not applicable

| Component          | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Sodium borohydride | 16940-66-2 | -                                                                   | -                                                                             | -                                                                                                     |

Seveso III Directive (2012/18/EC)

| Component          | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Sodium borohydride | 16940-66-2 | Not applicable                                                                            | Not applicable                                                                           |

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 94/33/EC on the protection of young people at work

Take note of Dir 92/85/EC on the protection of pregnant and breastfeeding women at work

## National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component          | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|--------------------|---------------------------------------|-------------------------|
| Sodium borohydride | WGK2                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H260 - In contact with water releases flammable gases which may ignite spontaneously

H301 - Toxic if swallowed

H314 - Causes severe skin burns and eye damage

# SAFETY DATA SHEET

Sodium borohydride caplets

Revision Date 06-Oct-2023

H318 - Causes serious eye damage  
H360FD - May damage fertility. May damage the unborn child  
EUH014 - Reacts violently with water

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Creation Date** 29-Jan-2010  
**Revision Date** 06-Oct-2023  
**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**